

Here's some documentation for users that I dug up in ancient files, today, 2011-04-07.

--Pete

**Doc: 'UN1' (UN = User Note)**

ZYKLUS MIDI PERFORMANCE SYSTEM

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Pete Kellock

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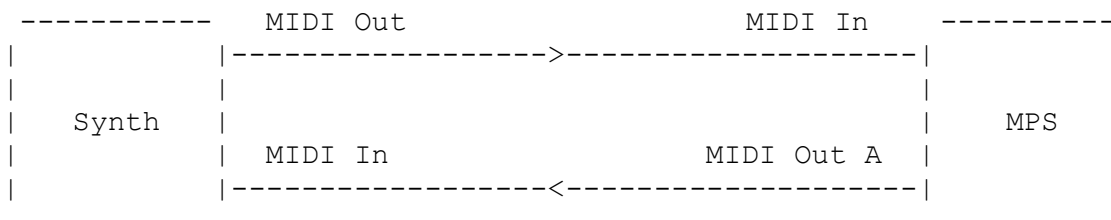
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| GETTING STARTED |  
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From first taking the machine out of its box, you can be up and running in 5 minutes if you follow the guidelines below. From there on, if you have difficulties in finding or understanding a feature, refer to the User Guide. It has an index and a 'Problems Page' for quick reference.

1. Take the MPS out of the box and check that the Voltage Selector on the back is set to your local voltage.
2. On the back of the MPS, connect the MIDI Input plus MIDI Output A to a synth (or other MIDI gear) as follows:



3. On the synth, set the transmit & receive channels to 1.
4. Plug in the mains and switch on. Adjust the display brightness & viewing angle by turning the 2 controls on the base of the unit. You should now see the following display:

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120  A1 A1 A1 A1 A1 A1 A1 A1 A1 A1 A1 A1      (Output A, channel 1)
120   T  T  T  T  T  T  T  T  T  T  T  T      ('Thru' in all 12
                                           positions)

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[ If you don't have the above display, the memory probably isn't blank so you should format it. The quick way to do this is to switch off the machine, hold down blue keys 1, 3 & 5 only, then switch on. BE VERY CAREFUL ANY TIME YOU USE THIS FEATURE - IT ERASES THE ENTIRE INTERNAL MEMORY WITHOUT WARNING YOU! ]

5. Press & release the ASSIGN SEQ key (4th black key at top), turn wheel to right and press the ENTER key to select ASSIGN-12-FREE. You will now have the following display:

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120  A1 A1 A1 A1 A1 A1 A1 A1 A1 A1 A1 A1      (Output A, channel 1)
120  01 02 03 04 05 06 07 08 09 10 11 12      (Seqs 1 to 12 assigned)

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You now have 12 free sequences and can record on any of them.

6. Press the REC/EDIT key then Control Key 1 (the 1st blue key). The display will change completely - you are now in step-time record on sequence 1. Play a few notes near the bottom of the keyboard and you will see them appear at the right of the display.
7. Now record a few notes into sequence 2. To do this, press REC/EDIT then Control Key 2 and play in the notes. Use notes near the top of the keyboard so that they're clearly distinguishable from sequence 1. Notice that you're still hearing seq 1 while you're recording seq 2 . If you don't want this, press Control Key 1 again and it's LED will go off (to show that you're not monitoring it any more).
8. Press PLAY then RUN/STOP. You can now replay your 2 sequences, triggering and controlling them in hundreds of different ways. The

current triggering mode depends on the 12 grey keys at the right of the machine and the STEP key at the left - these 13 keys are known as the 'Trigger Profile' keys.

9. Start with all 13 Trigger Profile keys off. Press the blue Control Keys 1 and 2 to hear your sequences. Switch these keys on & off a few times - notice how you have independent control over each sequence and that they each start from the beginning when you activate them.
10. Switch off both sequences, press the EXT TRIG key at the right and hit Control Key 1 again. Notice that it now flashes to tell you it's ready. Hit a note on the synth and the sequence should be triggered (if not, check the connections and make sure that the RUN/STOP key is still on). Experiment with the following cases:
  - a) Hit a note on the synth, release it, then hit another. Note how the sequence is transposed and re-triggered each time.
  - b) Hold down a note on the synth, then press another while keeping it held. Note that 2 transpositions of the sequence are now running in parallel and the sequence is not re-triggered till you release all notes and press a new one. You can have up to 8 transpositions running in parallel - ie up to 8 'Control Pitches' for each sequence.
  - c) Try hitting the synth notes with different touches (hard & soft). If you're using a velocity-sensitive synth with a velocity-sensitive sound you'll hear that the velocity is being controlled from the keyboard.
  - d) While seq 1 is running, press Control Key No 2. You can now trigger seq 2 from the synth keyboard. Note that whatever Control Pitch(es) you set for seq 1 are now fixed or 'pitch-locked'. (Normally a sequence becomes pitch-locked if it is running when you press the Control Key for another sequence.)
  - e) You can stop or 'kill' either sequence by pressing its Control Key while it is running. You can then set it ready (flashing) by pressing it again - and so on.
11. Kill both sequences and switch on the STEP key at the left side of the MPS.

You can now single-step through the seqs from the blue Control Keys or from the synth keyboard (if EXT TRIG is on). Note that if EXT TRIG is off and you want to kill a sequence, you must release STEP before pressing the Control Key.

12. Put on the MOM (Momentary) key and experiment with various combinations of MOM, EXT TRIG and STEP. The effect of the MOM key should soon become clear.

You have now covered maybe 2% of what the MPS can do, but hopefully you can begin to see the possibilities for triggering & controlling sequences in real time - ie PERFORMING the sequences. This is essentially what the MPS is all about, but many of the possibilities which arise from this basic idea are surprising. The MPS is as much a 12-channel MIDI effects unit or a 12-channel MIDI control station as it is a sequencer.

To explore forward from this point it is probably best to experiment with the features introduced above but using more sequences. If you want to record some sequences in real-time, just hit RUN while in step-time record mode. (You should normally start with a free sequence - attempting to play over an existing sequence causes a punch-in, NOT an overdub.) You set the loop point in real-time record mode by hitting ENTER or RUN/STOP. To start recording a different sequence you then press another Control Key.

You may find it useful to go straight to User Note 2 at this point - it gives various tips which will allow you to get on faster.



Doc: 'UN2'

ZYKLUS MIDI PERFORMANCE SYSTEM

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4/2/88  
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| MISC TIPS |  
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If you're a beginner or near-beginner with the MPS, the following tips may help:

1. If nothing happens (eg no sequences run) when you press the blue Control Keys, check the following:
  - that you are in Play mode (PLAY is on, REC/EDIT & PERF are off)
  - that RUN is on
  - that the sequences assigned to the Control Keys have some notes in them. (You can check by going into step-time record mode.)
  - this is less likely, but you might have somehow selected EXTERNAL CLOCK. If unsure, press TEMPO and look at the left of the display. If it says EXT CLK you will have to reselect INTERNAL CLOCK under the black GENERAL OPTIONS key.
2. If you have recorded some sequences, are now in Play mode, and your sequences seem to have mysteriously vanished, you have probably changed Configuration. To restore your previous Config, press the CONFIG key, then keep pressing ENTER until you see your original 12 sequence numbers.
3. If you can't switch the Control Keys off in Play mode, you probably have the STEP Trigger Profile key on. This is perfectly normal healthy MPS behaviour. Switch off STEP and you will be able to switch off the Control Keys.
4. You cannot go from RECORD to PLAY modes or vice versa when the machine is

running. Similarly the 8 black 'Function' keys at the top have no effect when the machine is running.

5. In step-time record mode you change step size, enter rests, etc by HOLDING DOWN the STEP key and pressing the Control Keys - see the legending above the Control Keys for details.
6. To record chords in step time, you hold down at least one note on the synth and press the other notes in the chord (up to 8). The time only moves forward by a step when you release ALL notes on the synth.
7. In step-time record mode, the Start-Of-Sequence is shown by a > symbol and the End-Of-Sequence by a < symbol. If the display shows > < you are on a free sequence.
8. When you set a loop point in real-time record mode (by hitting ENTER or the ENTER footswitch), it sets the loop point TO THE NEAREST BEAT, so you have to press it quite accurately. However, if you don't get it right, you can press STOP, move forward to the End-Of-Sequence marker (which is the loop point), hold down the MOVE key, and wheel the marker to the time you want it at - then press RUN. Sounds tedious... but it's actually very easy when you get used to it.
9. To delete an unwanted sequence, do the following:
  - ... in PLAY mode: Hold down DELETE and press a Control Key which has the desired sequence assigned to it.
  - ... in step-time RECORD mode: Hold down ENTER and press DELETE.

What advantage does the MPS give you compared to conventional sequencers?

Most hardware and software sequencers are fine if a) you have a pretty clear idea what you want to do before you start recording and b) you want to end up with a piece of music in a single fixed final form. They are not so friendly if you are starting to build up a song from a couple of basic ideas or if you want to be able to structure the basic material for a song in different ways. The MPS gives you a lot of this flexibility and additionally opens up a whole range of new effects.

Users seem to have identified 4 main types of application where the MPS has the edge or is at least a useful addition to conventional sequencers:

1. Creative tool. The MPS lets you record basic musical ideas very much as you would on a conventional sequencer, then allows you to trigger these in any order and combination and at different transpositions to form longer sections. The 'basic ideas' can be anything from short phrases, melodies, rhythms, single chords, etc, up to entire tracks or sections of the music. Typically you would use a mixture of these - so you might start from a repeating 2-bar drum pattern, bring in an 8-bar bass track, then fire off various short phrases/rhythms/chords/etc in real-time over the top. As you do this, everything that's happening can be recorded.

This way of working applies as much to getting a groove going for a pop or hip-hop track as it does to working up a jingle from a short hook line - or for that matter to the classical concept of developing a composition from a single musical idea.

2. Production/remix tool. The same type of principle applies to building longer sections into an overall song structure. This can be controlled spontaneously in real-time then edited to give various finished versions - the ideal tool for 12" remixes! When several people build complete pieces from the same material the results can be drastically different.
3. Live performance sequencer. The MPS allows the musician to choose to what extent the shape of a song is predetermined before the gig and to what extent it is controlled live in real-time. Like any sequencer it will allow entire songs to be recorded in fixed form; at the other extreme a piece can be improvised live on the MPS by spontaneously triggering individual tracks. A useful mid course is to program the song as a number of sections and to exercise live control over the transitions from one section to the next, leaving each section to loop as many times as required - so if the guitarist decides to do a 20-minute solo, the sequencer can be made to wait before going on to the next section (or it can be moved on to the next section when the rest of the band has had enough!).
4. Effects generator / 'sound processor'. MIDI triggering allows a whole range of effects based on the interaction of sequence and live data. Here are examples:
  - a) Chord and arpeggio effects triggered from MIDI. A unique feature of the MPS is 'hold-at-end' which allows a track to run through then hold the final note or chord as long as any notes are held. If the first part of the track consists of very rapid notes, they are heard as part of the timbre rather than as a sequence of notes - but obviously the effect IS sequence data and hence totally editable.



b) Tracks consisting of a repeated note can be used for superposing various effects on live MIDI input. For example, a track consisting of a rhythmic pattern 'tapped out' on a single note can be used to superimpose this rhythm on live playing. A track consisting of a single note repeated several times with decaying velocity introduces a 'delay line' effect when triggered from live MIDI data. A track consisting of a fast repeating note effectively 'strums' live input, etc.

As each of the above effects takes one of the 12 tracks of the MPS, any combination of 12 can be triggered simultaneously. One of our standard demo examples is an extreme case of this where 8 tracks are used to simulate a huge orchestral chord - the tracks introduce various delay effects (because the perceived 'size' of an orchestral chord arises partly from the fact that real players are not perfectly in time) plus octave transpositions, string tremolando effects, harp arpeggios, etc. The resulting sound (which we demo using only a TX81Z and an MT32) is massive - but unlike a sampled orchestral effect, it doesn't raise loop point issues and is quickly editable as MIDI data.

These are the main MPS applications discovered so far. Even these do not fully utilise its capabilities, so there may still be several applications waiting to be discovered.



*Doc: 'EXP'*

Using the Zyklus MIDI PERFORMANCE SYSTEM as a tool for Musical Experimentation

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8/2/88

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This document covers some aspects of the MPS which may interest any musicians involved in 'musical experimentation'. This isn't just intended to include what's usually termed 'experimental music' - the ideas covered here could be as useful in some pop/rock/jazz/new age music as in avant-garde/aleatory/ /minimalist/serial music. The main point is that the MPS allows rapid musical experimentation within chosen levels of predictability - not through setting up mathematical probabilities or compositional algorithms, but by the relatively straightforward principle that recorded sequences can be independently controlled and manipulated in real-time during playback. A secondary point is that the MPS is an efficient tool for recording and editing sequences where a high degree of precision is required - again this could apply as much to experimenting with the 'feel' of a drum part as entering a mathematically-defined serial composition.

#### SYSTEM OVERVIEW

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Although the MPS has many of the features of a conventional sequencer, the principle of independent real-time control over recorded sequences leads to many possibilities which are not always immediately obvious and are not generally associated with sequencers. The MPS is designed to make a large number of these possibilities rapidly accessible, both for live use and to

facilitate experimentation in a studio or similar environment. Its overall system architecture and its manner of operation are therefore significantly different from conventional sequencers, particularly in its Play modes - sequences can be PERFORMED rather than just replayed. The main elements of the design which make this possible are as follows:

1. Up to 12 polyphonic sequences can be triggered independently in real time very much as if they were single notes or sound events. Triggering can be from the front panel, from an external MIDI control source (typically a synthesiser keyboard), from a footswitch or from a gate signal. The relative synchronisation of the sequences is controlled by several options. For example, a sequence can start immediately it is triggered, regardless of any other sequences running, or it can be forced into a certain level of 'Relative Quantisation' (say to the nearest semiquaver relative to another sequence), or when triggered it can pick up in the middle as if it had been running silently in the background. Sequences can also be single-stepped from the control source.
2. When triggered from a MIDI source, each sequence can run simultaneously at up to 8 transpositions, and the 8 'Control Pitches' can be different for each of the 12 sequences. The resulting pitches are a real-time matrix multiplication of the sequence pitches and the Control Pitches. For example, if the sequence had a C major triad at a certain point in time and the Control Pitches were A and F#, the resultant pitches would be  $(A + C\# + E) + (F\# + A\# + C\#)$ . Because both the sequence and the Control Pitches are functions which vary in time, the harmonic, rhythmic and harmonic/rhythmic information can be spread between the sequence and the 'live' data to any chosen degree. To give 2 extreme examples:

- a) A 12-voice polyphonic work could be recorded as 12 complete sequences routed to different timbres on 12 MIDI channels. The control pitches could then be used to 'arrange' these parts, experimenting with various octave offsets or doublings applied to each part.
  - b) Rhythmic patterns can be recorded as sequences on a single pitch. These patterns can then be superimposed on live playing.
3. Each Control Pitch of each sequence has a MIDI note velocity value associated with it. These 'live' velocity values can be averaged with the sequence velocity values in various proportions to produce the resultant velocities.
  4. The way in which control pitches are assigned to the 12 sequences is selectable from a large number of options. For example, the control pitches for certain sequences can be 'frozen' while others remain active, or incoming pitches can be routed in a cyclic manner to a combination of active sequences.
  5. A second level of transposition is provided - this causes the Control Pitches themselves to be transposed. Normally all active sequences are affected, but an option is provided for inhibiting transposition of certain sequences.
  6. The way in which the controls respond and interact is also selectable. For example, if a MIDI keyboard is being used, sequences can be selected to run only while notes are held on the keyboard, or to continue running at a given set of pitches until a new note or set of notes are pressed.

7. Most of the options described above are selected by single key presses and can be altered in real time during replay. To be more precise, the control options or 'Trigger Profile' is defined by the state of 13 on/off parameters which can be set in any combination, so in theory there are 8192 ( $2$  to the  $13$ ) possible Profiles! In practice there are probably several hundred useful distinct modes. These can be mixed in various ways - for example some sequences can be running while others are being single-stepped.
8. Sequences can also be assigned one of 3 'Repeat Modes' : Repeat, Single-Shot or Hold-At-End. The first 2 are common to most sequencers, but Hold-At-End is a novel feature allowing a sequence to run through and come to rest on its final note or chord. In a typical case a sequence will consist of some rapid notes ending on a note or chord.
9. Sequences can be set to any length, definable to within a single clock pulse ( $1/96$  of a beat). All sequences run independently, so any which are set to Repeat will loop independently of others.
10. Because MPS sequences are often small elements of an overall musical structure, it is useful to have the freedom to 'pick & mix' existing sequences, sometimes using the same sequence in several musical contexts. The MPS architecture facilitates this by allowing any combination of 12 sequences to be assigned to the 12 'positions' on the front panel for immediate use. A set of assignments plus other associated data forms a 'Configuration'. There are 24 Configs, any of which can be rapidly selected - typically by a single key or footswitch press. The associated data includes 12 MIDI destinations (ie bus & channel number)

for the sequences, plus a large number of optional parameters - eg MIDI mode & program changes, on/off startup states for the 12 assigned sequences, tempo values, etc. All such parameters have a 'no change' option, so Configs can be used for subtle or drastic changes in the system setup.

11. The MPS also allows 12 'Performances' to be recorded, edited and replayed. In a Performance, all actions are recorded: identity & timing of sequence triggerings, MIDI events received, tempo & Config changes, etc
12. The MPS has several additional features generally associated with 'MIDI Utility Boxes' rather than sequencers. For example, it allows complex MIDI Thru routing to multiple destinations, keyboard zoning, step-through of MIDI program changes, etc. In combination with the sequence control options, this allows the MPS to be used as a central MIDI control station.

#### EXPERIMENTAL TECHNIQUES/EXAMPLES

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1. As stated above, repeating sequences have independent loop points. In a typical rock example, this means one sequence can be a bar long, another 8 bars and another 32 bars. More unusually, one sequence can be 4 beats long and another 5 beats - obviously this causes a gradually changing rhythmic pattern which repeats every 20 beats. A more extreme example is to mix sequences which differ in length by one or more clock pulses to produce phasing effects. The clock resolution is a 96th of a beat, so a 4 beat sequence and a 4 beat + 1 clock sequence will produce a pattern which repeats after 384 times through the longer sequence - fine enough

resolution for most phasing effects.

Note that a 'beat' defaults to a value of 1 crotchet, but the time signature can actually be set anywhere in the range 1/2 to 32/8. Although this may seem slightly limiting in the denominator value, it is unlikely to present a problem as the time signature only affects metronome speed and bar/beat display while recording - in general the MPS makes little use of, and is not limited by, concepts of bar or beat.

2. In assigning sequences to the 12 positions on the front panel, the same sequence can be assigned to several positions. This allows complex imitative textures to be built up where relative timing can be free/quantised/etc and different transpositions & velocities can be applied to each part. This can be improvised in real-time or tightly controlled by setting the trigger points to single clock-pulse resolution in a Performance (see 10 above for definition of a Performance).
3. As sequences can be triggered by MIDI note on/off data, many possibilities exist for triggering sequences from other sequences. The resultant patterns can be tightly controlled or pseudo-random, depending on the relative lengths of the sequences, the times between events, etc. The controlling sequence could be a Performance (essentially a Performance IS a sequence with some additional types of event, and is recorded in a very similar manner). Alternatively one of the MPS outputs can be looped back to the input and some sequences used to trigger others, or sequences used to transpose themselves. Using external equipment many other possibilities exist - for example triggering from an external sequencer or merging looped-back sequences with live data.

In all these cases different sequences can be triggered from different MIDI

ranges and/or respond cyclically to MIDI events. It is also possible to impose the rhythm of one sequence on another by disabling transposition on the controlled sequence then single-stepping it from the controlling sequence.

4. The 'Hold-At-End' feature described above can be used to produce quasi-timbral and ensemble effects. The most obvious technique is to use very rapid introductory notes to produce 'chiff' effects before the final note or chord. A less obvious technique is to record several single-note sequences with varied initial delays of a few clock pulses, route these to different MIDI destinations, then trigger them simultaneously from live MIDI data - the effect can be similar to ADT (Automatic Double Tracking, a standard studio technique) or in more complex cases can simulate the effect of subtle timing errors when several musicians play 'in unison'.
5. Other delay effects can be created using simple sequences. For example, a sequence can consist of a repeated note with decaying velocity; when this is triggered from a MIDI source and routed to an appropriate velocity-sensitive sound, the effect can be similar to a delay line with feedback. However, unlike the delay line, each repetition can change octave or undergo any other pitch or velocity shift required.
6. The MPS is specially suited to creating textural effects based on rapid internal movement: strumming & arpeggiation effects, rolls etc. These are based on rapid repeating sequences triggered from a MIDI source. For example, a simple sequence based on octaves can be used to produce a halo of rapid movement round a melodic line.



7. Interesting semi-determinate effects can be produced when the MPS is used to control a device which has several sounds each allocated to different MIDI pitches, such as the majority of drum and percussion sound generators. When a sequence routed to such a device is transposed, the rhythm is preserved but the sound assignments are changed. Use of multiple Control Pitches can cause the rhythm to be doubled on several sounds. Because the MPS allows individual sequences to be triggered/transposed independently AND has a second level allowing overall transposition, subtle or drastic changes can be effected in response to single MIDI note events. With a little experimentation this can be a powerful technique for live improvisation.

8. As stated above, the MPS is an efficient tool for recording and editing sequences when high precision is required. Although this is conventional sequencer territory, the MPS has several advantages in terms of quick operation. For example, when entering notes in 'step time', a wide selection of step sizes are available and the on-time of each step can be set from 0% to 255% of the step size, the latter allowing overlapping notes to be entered very easily. Chords are entered by holding down several notes - time moves forward to the next step only when all notes are released. However, the most powerful features are the single-event editing controls:

DELETE: Single note delete

VAL:           Used to view & alter value of a MIDI event. In the case of note events, this refers to the velocity value.

LENGTH: Used to view & alter note on-time.

MOVE:    Used to move a single event in time by units of one clock pulse. The timing of other events is not affected.

SLIDE:   Used to move an event AND ALL FOLLOWING EVENTS by units of one clock pulse - ie to time-slide sections of a sequence.

All these controls are based on single-keystroke operation. For example, to move an event, the cursor is positioned on the event, MOVE is held down and the wheel is turned to move it; a continuous read-out of time is given. On releasing MOVE the note is at the new time position. (The operation typically takes 2 or 3 seconds to perform.)

Together these controls allow note on/off and other MIDI events to be positioned accurately with a resolution of 1/96 of a beat. A possible objection is that 1/96 of a beat is not sufficiently accurate. However, this represents resolution to approx 2.5 mS at maximum tempo, which is accurate enough for most purposes, firstly because such delays are detectable only in the case of percussive or other sounds with fast attack transients (and normally only when such sounds are nominally simultaneous - a 2.5 mS shift is seldom detectable in a sequential series of events), and secondly because this is within the region of MIDI and software delays. (For comparison, it takes approx 1mS to transmit a Note On or Note Off event over a MIDI link.)

Hopefully the above gives some idea of the potential for experimentation with the MPS. Because it is innovative and generally 'open-ended' in design, there are no doubt many other possibilities - particularly in the area of Trigger Profile, where the sheer number of combinations means that many are as yet untried. We welcome hearing from any users who discover useful new techniques and are willing to share the information. We also welcome suggestions for improvements and upgrades - sometimes these turn out to be easily implementable in software (though often not, of course!) so even changes which are suggested by only a few people will at least be seriously considered.



Doc: 'SEP'

## The Art of Sequencing According to Zyklus

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(Article written for SE + P)

Pete Kellock

29/4/88

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Who says MIDI's a good thing? I seem to remember a time in the dim & distant past (5 years ago?) when there were electronic instruments on the one hand and pro audio gear on the other. You dealt with synths & sequencers or you dealt with desks and outboard, and music software was an area of academic research as relevant to most people in the music industry as molecular biology. We thought life was complicated, but little did we know...

Look what MIDI's done for us. Lots of new possibilities. So many possibilities that we spend half our time trying to figure out what various bits of equipment/software can and can't do, how easily they let you do it, how they can be interfaced with other gear, what the software upgrade paths might be, etc, etc. Result - boggled brain cells and less time to think about music.

In designing the Zyklus MIDI PERFORMANCE SYSTEM (MPS), it could be argued that we have simply assisted this trend. We have produced a system that creates a whole new set of possibilities which can't be neatly categorised and are fairly difficult to describe in words. To make matters even worse, we seem to have opened up several entirely new directions which may breed whole FAMILIES of new products and new possibilities. Aaaagh!

Why have we done so? To get to the point - because we think that conventional sequencers are too rigid and inflexible for many purposes. They are fine if a) you have a pretty clear idea what you want to do before you start recording and b) you want to end up with a piece of music in a single fixed final form. They are not so friendly if you are starting to build up a song from a couple of basic ideas (such as a rhythm pattern and a bit of melody), or if you want to be able to structure the basic material for a song in different ways (eg for a remix or to allow a bit of flexibility during a live gig).

The MPS gives you a lot of this flexibility and opens up a range of new effects. All the possibilities arise from the one central idea of real-time sequence control during playback. Contrary to some widely-held beliefs they have nothing to do with algorithmic composition, statistically-controlled random variations or molecular biology. In other words the MPS lets you record basic musical ideas very much as you would on a conventional sequencer, then allows you to trigger these in any order and combination and at different transpositions to form a complete piece of music. The 'basic ideas' can be anything from single phrases, melodies, rhythms, chords, etc, up to entire tracks or sections of the music. Typically you would use a mixture of these - so for example you might start from a repeating 8-bar bass track, bring in a 4-bar drum pattern, then fire off various short phrases/rhythms/chords/etc in real-time over the top. As you do this, everything that's happening can be recorded.

The real power and innovation of the MPS lies in the range of triggering options or 'Trigger Profiles'. There are a vast number of these options covering aspects such as the trigger source (front panel of the MPS, external MIDI source, or footswitch), the relative timing of tracks when they are triggered, and the way pitches are assigned to tracks when triggering from MIDI. (Note: we normally use the term 'sequence' to mean the basic triggerable chunks of MIDI data, but I will use the term 'track' in this article as it probably conforms more closely to standard sequencer terminology - but bear in mind that MPS 'tracks' are much more independent than the tracks of a normal sequencer.)

The possibilities for MIDI triggering are probably the most complex aspect of the MPS. To say that the MPS allows sequences to be transposed is rather like saying that a Synclavier allows you to play tunes on a keyboard. In both cases the misleading simplicity of the statement can easily be demonstrated but is a little more difficult to describe. What the MPS actually does is a real-time matrix multiplication of the pitch/velocity data in a track and the pitch/velocity data arriving on the MIDI input - which unfortunately is another fairly useless description unless you have a degree in Applied Maths (and possibly even if you do). To get closer to the idea, think in terms of tracks being triggered from a keyboard as if they were single notes. The track will be transposed by the note(s) played on the keyboard and its velocity will depend on the velocities of the notes played (you can mix the original velocities in the track with the 'live' velocity in various proportions). Each of the 12 tracks can have up to 8 transpositions running at the same time, each with its own velocity, and the 8 transpositions can be different for each track. The timing of the track relative to other tracks at the point of triggering also depends on several options. All of these options (in fact all parameters of the Trigger Profile) can be altered while other tracks are running.

What use is all this in practice? So far, we (and other users of the MPS) have identified 4 main types of application where the MPS is a superior alternative or useful addition to existing software/hardware sequencers:

1. Creative tool. As I've mentioned already, the MPS allows you to experiment rapidly with basic ideas and develop them into longer sections. This applies as much to getting a groove going for a pop or hip-hop track as it does to working up a jingle from a short hook line - or for that matter to the classical concept of developing a composition from a single musical idea.
2. Production/remix tool. The same principle of combining musical elements in different ways applies to building longer sections into an overall song structure. This can all be controlled spontaneously in real-time then edited to give various finished versions - the ideal tool for 12" remixes? When several people build complete pieces from the same material the results can be drastically different.
3. Live performance sequencer. The MPS allows the musician to choose to what extent the shape of a song is predetermined before the gig and to what extent it is controlled live in real-time. Like any sequencer it will allow entire songs to be recorded in fixed form allowing most of the band to nip offstage for a pint; at the other extreme a piece could be improvised live on the MPS by spontaneously triggering individual tracks. A useful mid course is to program the song as a number of sections and to exercise live control over the transitions from one

section to the next, leaving each section to loop as many times as required - so that the keyboards player has the freedom to do a 20-minute solo if he wants. (A software upgrade will allow pacification of the rest of the band via SysEx messages.)

4. Effects generator / 'sound processor'. MIDI triggering allows a whole range of effects based on the interaction of sequence and live data. Here are 2 types of effect we have explored in some depth:

- a) Chord and arpeggio effects triggered from MIDI. A unique feature of the MPS is 'hold-at-end' which allows a track to run through then hold the final note or chord as long as any notes are held. If the first part of the track consists of very rapid notes, they are heard as part of the timbre rather than as a sequence of notes - but obviously the effect IS sequence data and hence totally editable.
- b) Tracks consisting of a repeated note can be used for superposing various effects on live MIDI input. For example, a track consisting of a rhythmic pattern 'tapped out' on a single note can be used to superimpose this rhythm on live playing. A track consisting of a single note repeated several times with decaying velocity introduces a 'delay line' effect when triggered from live MIDI data. A track consisting of a fast repeating note effectively 'strums' live input, etc.

As each of the above effects effectively takes one of the 12 tracks of the MPS, any combination of 12 can be triggered simultaneously. One of our demo examples is an extreme case of this where 8 tracks are used to simulate a huge orchestral chord - the tracks introduce various delay effects (because the perceived 'size' of an orchestral chord arises partly from the fact that real players are not perfectly in time) plus octave transpositions, string tremolando effects, harp arpeggios, etc. The resulting sound (which we demo using only a TX81Z and an MT32) is massive - but unlike a sampled orchestral effect, it doesn't raise loop point issues and is again totally editable as MIDI data.

In all these cases, options are provided to trigger different tracks from different keyboard zones, or to trigger them cyclically (so the first note triggers track 1, the next track 2, etc). These open up yet more possibilities for interactive performance effects.

Hopefully this gives some insight into what the MPS can do and how it does it. We have certainly created new possibilities and in so doing have inevitably made the 'MIDI jungle' an even more confusing place. However, the MPS is in many ways a very quick system to operate - for example you can start from blank memory, enter several tracks and be triggering them from MIDI within 5 minutes. Our design philosophy is that flexibility and open-endedness should be the first goal, but 'musician-friendliness' should be a close second.



Doc: 'DEMO'

MPS PRESENTATION WITH ALEX: GIGSOUNDS 3/2/88  
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PART 1  
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Hi. I'm Pete Kellock from Zyklus. I'm one of the designers of the MIDI PERFORMANCE SYSTEM we're about to show you. I'd like to introduce Alex White who's NOT from Zyklus but who's been spending a lot of time with the MPS recently and probably knows it better than we do by now.

OK, what we're going to try to do just now is give an answer to that famous question "What the Hell's a MIDI Performance System?". Well... there are several answers to this, and some of them are even repeatable.

\*\*\*

For example - answer number 1 - it's a sequencer. Here are some sequences "we prepared earlier" as they say (I'll come back to how you record them later).

- \* CONFIG 1. "PETE - DEMO PIECE"
- \* TP: ALIGN
- \* 12 'TRACK'-TYPE SEQS. SWITCH A COUPLE ON & OFF (LIKE MONITOR SWITCHES)

(I think you can see there that you basically have 12 tracks and Alex was muting them off and on from the 12 blue control keys.)

\*\*\*

So... it's a sequencer, but then again the sequences can be short phrases which you can "play" from the front panel using the 12 blue control keys along the front - almost as if the MPS was a musical instrument in its own right:

- \* CONFIG 2. "THRILLER"
- \* TP: GROUP + EXCL
- \* IMPROVISE FROM CONTROL KEYS: CHANGE TP & DO SOME MOMENTARY STUFF

(If any of you have used lighting controllers you might notice some similarity in the way Alex was using the machine there - in fact we got some of the ideas for that kind of live control from stage laser control systems.)

\*\*\*

Then again, it can be a MIDI Control Station:

- \* TURN DOWN VOLUME ON DX7
- \* CONFIG 3. "CONTROL STATION"
- \* TP: EXCL

For example it can handle routing to different MIDI destinations and will send out MIDI program changes to set up the sounds on each destination:

- \* a. DIFFERENT DESTS

... or it can route you to several destinations in parallel:

- \* b. MULTIPLE DESTS IN PARALLEL (SEQS 6 - 9)

... or you can have some wierd effects such as cycling between destinations:

- \* c. CYCLING

... it can also zone the keyboard

- \* d. GROUP - ZONING (SEQS 4 + 8 + 11)

\*\*\*

The MPS can also be used as a keyboard or MIDI effects processor. For example, playing chords from single notes:

- \* CONFIG 5. "HOLD-AT-END & SINGLE-SHOT EXAMPLES"

- \* TP: EXT + MOM + EXCL

- \* PLAY 1 & 2 (HOLD-AT-END)

... or getting arpeggiator-type effects from single notes

- \* CONFIG 5.

- \* PLAY 3, 4, 5 (SINGLE-SHOT)

- \* PLAY 6 (REPEAT)

... or you can have effects which are a combination of arpeggio & chord

- \* CONFIG 5.

- \* PLAY 7 - 10 (HAE)

- \* NB: YOU MAY HAVE TO PLAY 9 A FEW TIMES (IT CONFUSES THE MT32 VOICE ASSIGN)

... or you could use it to superimpose a rhythm or a sequence onto live playing:

- \* CONFIG 6.

- \* a. PLAY 1, 2 & 3 INDIVIDUALLY - SIMPLE CHORDS (3 - 4 NOTES)

(You can also combine these rhythms:)

- \* b. PLAY 1 + 2 + 3 TOGETHER - MINOR 9TH CHORDS?

\*\*\*

With some of these keyboard effects you can run them so fast that they become part of the sound. For example, here's a sound on its own:

\* CONFIG 6.

\* a. PLAY 4 - FEW NOTES, MELODIC

... now you could use the MPS to 'strum' that sound

\* b. PLAY 5 - SAME FEW NOTES

Here's another sound:

\* c. PLAY 6 - FEW NOTES, MELODIC, SLOW

... and you could use the MPS to give you some very fast notes at the start, producing a sort of 'chiff' effect:

\* d. PLAY 7 - SAME FEW NOTES

... and you could combine these effects:

\* e. PLAY 5 + 7 - SAME IDEA

... or you could combine these with the rhythm we had above:

\* f. PLAY 1 + 2 + 3 + 5 + 7 - HOLD SINGLE LONG NOTES & MAYBE A FIFTH

\*\*\*

So, to answer "What the Hell's a MIDI PERFORMANCE SYSTEM" ... it's a sequencer, but it's a sequencer that's almost a musical instrument in its own right, it's a MIDI Control Station, it's a MIDI effects processor and it can even be a sound processor.

What you may be wondering at this point is whether there's any connection between all these effects. Well, as you may have guessed - just about everything you heard there was based on the one central principle of sequence manipulation. Those were all sequences played in different ways. (The only exception being when we were simply passing data straight through, for example in some of the Control Station applications.)

So what the MPS gives you is a whole range of ways of CONTROLLING sequences - you can control them from the front panel of the MPS or from a keyboard (or any other MIDI source for that matter) or from a footswitch. When you control them from MIDI they take the pitch or pitches of the notes you play - you can have each sequence running at up to 8 different pitches (and the sequences themselves can be up to 20-note polyphonic, and also you can have a different set of 8 pitches for each of your current 12 sequences - assuming you've got enough synths to handle all that!)



So the MPS is a SYSTEM for PERFORMING sequences (or MIDI data) - that's why we've called it a MIDI PERFORMANCE SYSTEM and not a sequencer.

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PART 2

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OK, now you've seen a few of the ways you can make use of sequences, let's go back to square one - ie how do you get the sequences in there in the first place? In this respect it's pretty much what you'd expect of a dedicated sequencer with most of the features you'll find on any other good sequencer plus a few extras.

- \* PULL OUT CARTRIDGE
- \* SWITCH OFF. HOLD DOWN CONTROL KEYS 1,3 & 5 AND SWITCH ON AGAIN
- \* TURN DOWN VOLUME SO I CAN SPEAK OVER MUSIC

To show you just how quickly you can get started on it, we've cleared the memory. Right - the machine is now in exactly the state it's in when you first take it out of the box. The first thing you do is give yourself 12 unused or 'free' sequences to start recording on. What you're actually doing is assigning the first 12 seqs to the 12 control keys. The machine can hold 99 seqs in its internal memory and another 99 on cartridge; you can have any combination of 12 of these available at any one time.

- \* ( WHILE I'M SAYING THAT LOT DO ASSIGN-12-FREE )

You can now build up sequences very quickly in step-time or real time. To illustrate this, Alex will now enter some sequences in step time then some in real time. To select the seq to record on, you simply press REC/EDIT then one of the Control Keys.

- \* RECORD 2 IN STEP TIME & 2 IN REAL TIME

To mention just a couple of points about that: To enter chords in step time you just hold down all the notes of the chord; when you release all notes it steps forward. The other point is that the loop points of all the sequences can be different.

(As you can hear we've got everything on one sound at the moment - we're only using the DX7 .)

\*\*\*

We're now back into step time. We'll just quickly show you a couple of the editing features. These can be applied to seqs you recorded either in step or real-time - there is no difference. To keep it simple for now, Alex has just put in a 4-note sequence which sounds like this:

- \* ( ENTER SEQ 12 WHILE I'M SPEAKING )
- \* LET THEM HEAR IT RUNNING

Now let's make the last note louder. You just move to the last note, hold down the VALUE key and it tells you what the velocity of that note is. To change it you just turn the wheel & release the key.

\* ( ETC )  
\* LET THEM HEAR IT RUNNING

Now let's move the last note. Again you move to it, hold down MOVE, turn the wheel and let go.

\* ( ETC )  
\* LET THEM HEAR IT RUNNING

\* ( WHILE I'M SAYING THE NEXT BIT, START ENTERING THE RHYTHMIC SEQS 5/6/7 )

I think you can hear that pretty clearly. And I think you can also see that these editing features are very fast to use so you can quickly alter the feel of sequence - or it could be drum part. There are 5 which operate in a similar way - as well as VELOCITY & MOVE you have DELETE to remove notes & other events, LENGTH which allows you to alter the on-time of notes and SLIDE which lets you time-slide a whole section of a sequence (unlike MOVE which as you saw was only affecting one note).

\*\*\*

(Alex has just been entering a couple of other seqs we'll use in a minute.)

OK, we're now going back into PLAY mode, and we'll show you just a few of the options for manipulating these seqs we've recorded. Most of these options are selected using the 12 keys on the right plus the STEP key on the left. We call these the Trigger Profile keys. The idea is that you can alter the TP very quickly just by hitting these keys and you can change TP while seqs are running - changing it never affects seqs which are running already, it only sets it up ready for the next seq you want to control.

The simplest TP is with all these off. In this mode the seqs just start whenever you hit the Control Keys:

\* TP: ALL OFF. EXAMPLE  
\* TP: MOMENTARY. EXAMPLE  
\* TP: RESTART. EXAMPLE

\*\*\*

Before we illustrate the next option, we'll just let you hear the rhythmic seqs Alex was entering a minute ago. On their own these sound like this:

\* TP: RESTART

\* PLAY 5 + 6 + 7 DIRECTLY FROM FRONT PANEL

As you can hear, these are all based on C - just rhythms with some octaves. If we now put on EXT TRIG, we can control these from the keyboard and give them all different pitches, so you build up these rhythms into a sort of animated chord:

\* TP: RESTART + EXT

\* BUILD UP 5 + 6 + 7 TO FORM CHORD

You can now put on TRANSPOSE and transpose all of that:

\* TP: RESTART + TRANSPOSE

\* EXAMPLE

And we could now bring in the other seqs we recorded:

\* EXAMPLE

\* BRING SEQS 1 - 4 IN AND OUT

\*\*\*

I hope that gives some impression of just a few of the sequence control options. That's only scratching the surface. In theory there are over 8000 different possibilities for TP and you can change between any of them in real time while the machine is running. There are various other options you can set up when the machine isn't running, so the possibilities are immense - we'd love to show you them all, but it's only 2 hours to closing time!

However, I will just quickly mention a couple of other aspects of the machine.

< Explain Configs >

< Explain Perfs >

\* RECORD A VERY SHORT PERF & PLAY IT BACK

Once you've recorded a Perf like that, you can edit it in much the same way you edit a sequence - delete mistakes, correct bad timing, add things, etc.

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### PART 3

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\* WHILE I'M BABBLING, SET UP FOR PERFORMANCE OF ALEX DEMO

OK that's the MPS in a nutshell. In terms of hardware the whole system is designed to withstand everything it might come across in live performance or on the road. For example there is extensive mains filtering and transient suppression, the memory is battery-backed so if the power goes down you don't lose your data - in fact there are 2 batteries so if the first fails the

second will take over. We decided against using disk drives because they're relatively fragile and slow to load. Instead we use these cartridges which are also double battery-backed and as one reviewer said, "designed to withstand a nuclear explosion". (They might even stand up to a heavy metal gig.) You can actually run from the cartridge, so you don't even have to load it in.

In terms of connecting it up you have: 1 MIDI input and 4 MIDI outputs. There's SYNC IN & SYNC OUT with various clock speeds. Also, if you're using MIDI clocking the MPS supports MIDI Song Pointers. In terms of other I/O, there is a metronome output plus 3 footswitch inputs. We've also tried to support people who have older pre-MIDI gear or modular systems, so the footswitch inputs can take gate signals and there's a programmable gate output. There's even a special trigger input for very fast trigger signals.

To finish we'll play a complete piece written by Alex for the machine - the whole thing is stored as a Performance. After it I'll be here to try to answer any questions you have.

\* ALEX DEMO



Doc: 'DEMON'

MPS PRESENTATION WITH ALEX: GIGSOUNDS 3/2/88

-----

Intro: me & Alex

Setup

What the Hell's a MIDI PERF SYS? - Several possible answers

- 
1. Obvious answer:                 SEQUENCER.                         \*\* ALEX \*\*  
                                     (Muting on & off)
  
  2. Independent seq ctrl:     MUSICAL INSTRUMENT                         \*\* ALEX \*\*  
                                     (Lighting controllers)
  
  3. Also:                         CONTROL STATION  
    - routing + prog changes     \*\* ALEX \*\*  
    - multiple destinations       \*\* ALEX \*\*  
    - unusual effects - cycling\*\* ALEX \*\*  
    - zoning                       \*\* ALEX \*\*
  
  4. Also:                         KEYBOARD/MIDI EFFECTS PROCESSOR  
    - chords from single notes   \*\* ALEX \*\*  
    - arpegg.-type effects       \*\* ALEX \*\*  
    - arpegg + chord              \*\* ALEX \*\*  
    - add rhythm to live playing   \*\* ALEX \*\*  
    - combine rhythms              \*\* ALEX \*\*
  
  5. Run effects fast - becomes  
    part of sound:                SOUND PROCESSOR  
    - sound on its own            \*\* ALEX \*\*  
    - 'strummming' effect        \*\* ALEX \*\*  
    - sound on its own            \*\* ALEX \*\*  
    - 'chiff' effect              \*\* ALEX \*\*  
    - combine these               \*\* ALEX \*\*  
    - combine with rhythm         \*\* ALEX \*\*
- 

So answer to "What the Hell's MPS?" - 5 answers  
                                     - and can combine them

Any connection? Yes - all sequences controlled in different ways (except  
Thru connections).

MPS gives you control from - front panel

- MIDI
- footswitch
- gate

In case of MIDI control, seq transposed. (8 x 20 x 12 if enough synths!)

So MPS is system for performing MIDI data - a MIDI PERF. SYSTEM  
... much more flexible & creative than conventional sequencer.

-----

Back to square one - recording.

As you'd expect - all features of dedicated sequencer + few extra

To show how quick started, Alex has cleared memory.

Only thing before recording is to assign 12 free.

Can hold 99 internal + 99 cart. Any combination of 12 of these at one time.

-----

To start recording just press REC/EDIT + Ckey.

Can build up quickly in step-time or real-time - Alex will now do. \*\* ALEX \*\*

( Mention - quick build-up (Linn 9000 mode)

- different loop points
- 2 types of punch-in in real-time. Simplest just to play.
- Punch-ins don't cut notes
- in step time, can quickly change step size, enter rests/chords
- wide range of step sizes )

-----

Editing. Back to step-time.

... but seqs can be step-time or real-time

To keep simple - 4 note seq. Sounds like: \*\* ALEX \*\*

Velocity - last note louder. Play: \*\* ALEX \*\*

Move - shift last note. Play: \*\* ALEX \*\*

5 controls like that - designed for very fast use.

(VAL & MOVE) + DEL, LEN, SLIDE

Use to quickly correct errors or to alter feel - eg drum part

(Alex entering a couple of other seqs)



-----  
That's what MPS does. Briefly mention HARDWARE:

Designed for live use as well as studio

- engineered for life on the road & general rough treatment
- Not just in terms of casing, etc
- connectors - pro quality metal DIN connectors
- Also applies to electronics:
  - a) Mains filt./trans. suppr. - bad mains in stage envir. (Vangelis)
  - b) I/O overvoltage prot. - connect to nearly anything
- Decided against Disk Drive - relatively fragile (& SLOW)
- Instead all memory is battery-backed with 2 batteries
- No need to save data. Comes up exactly as on power-down
- Carts: Also double battery-backed. Reviewer: "Nuclear Explosion"
- Can also run from Cart so no need to load to internal

CONNECTING UP:

- 1 MIDI In & 4 MIDI Out. Can sync to MIDI & supports song pointers
- Sync In & Sync Out with various clock speeds
- Metronome Out
- 3 footswitch inputs. Can also use gate signals for pre-MIDI
- Gate Out
- Trigger In

-----  
To finish - Alex piece

Questions afterwards

Piece stored as Perf, so just hit Run.




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-----

The MIDI PERFORMANCE SYSTEM demo material comes on 2 cartridges. Cart 'A' consists of a single piece of music arranged as several configurations. Cart 'B' has a number of separate configurations intended to illustrate different uses of the MPS in a variety of musical contexts.

In order to ensure that the demo sequences are always assigned to appropriate voices, we use a standard MIDI setup based on 2 low-cost expander units: a Yamaha TX81Z and a Roland MT32. The only additional requirements are a master synth (preferably velocity-sensitive) and reverb (preferably MIDI-controlled).

#### MIDI Destinations

-----

A1            Master Synth  
B1 - B4        TX81Z  
C2 - C10      MT32. (C10 is drums)  
C16            Reverb (on same MIDI bus as MT32 using Thru connection)

#### Demo Cart 'A': Sounds

-----

Master Synth:    High marimba/block type sound. (2 octaves higher than normal)

TX81Z:            The following sounds must be assigned to Perfs:  
         Perf 22            A10/A10/A10/A31/ (MIDI chans 1 - 4)  
         Perf 23            B21/B5/B24/C4/            (ditto)  
         Perf 24            A31/D4/C4/A31/            (ditto)  
         These perfs are called up automatically by MIDI program changes.

MT32:            All sounds called up automatically by MIDI program changes

#### Demo Cart 'B': Sounds

-----

Master Synth:    Hard electric piano type sound (normal pitch)

TX81Z:            As above

MT32:            As above

#### Reverb

-----

The demo material sends out reverb program changes in the range 1 to 16 on destination C16. We normally use a MIDIVERB II with the program changes

cross-mapped to effect numbers as follows:

| Prog | FX | No | Description       |  | Prog | FX | No | Description        |
|------|----|----|-------------------|--|------|----|----|--------------------|
| 1    | 31 |    | Slow Gate 200mS   |  | 9    | 35 |    | Fast Gate 150mS    |
| 2    | 28 |    | Xlarge Warm 5S    |  | 10   | 5  |    | Med Bright 0.6S    |
| 3    | 21 |    | Large Warm 2.2S   |  | 11   | 32 |    | Slow Gate 250mS    |
| 4    | 22 |    | Large Warm 1.75S  |  | 12   | 27 |    | Large Bright 2.5S  |
| 5    | 13 |    | Large Warm 1.1S   |  | 13   | 8  |    | Medium Dark 0.6S   |
| 6    | 7  |    | Large Dark 1.0S   |  | 14   | 11 |    | Medium Bright 0.8S |
| 7    | 26 |    | Large Bright 2.4S |  | 15   | 22 |    | Large Warm 1.75S   |
| 8    | 3  |    | Small Bright 0.3S |  | 16   | 22 |    | Ditto              |

#### Note

-----

To send all MIDI program numbers for a Config, go to PLAY mode with the machine stopped, then hold down PLAY and press RUN - this restarts all seqs and sends all 24 MIDI prog changes for the current Config.

#### USING THE DEMO MATERIAL

-----

#### Cart 'A'

-----

This is a single piece of music written by session keyboards player Alex White. It can be heard in its entirety by running Performance 1. It uses Configs 1 to 9 - these can be used individually in Play Mode to illustrate different aspects of the Trigger Profile, etc.

#### Cart 'B'

-----

This consists of several distinct Configs put together by Pete Kellock of Zyklus. The Configs are arranged as follows:

1. General-purpose. Seq 1 is assigned to 2 Control Keys (7 & 8) which is useful for demonstrating the sequence synchronisation options (RESTART, ALIGN & REL QUANT).
2. Useful for illustrating asynchronous triggering from the Control Keys - almost anything works musically!
3. Using MPS as Control Station. Use Control Keys 4 - 12 to illustrate Thru routing to single & multiple destinations. Put on CYCLE to show cyclic routing and GROUP to show keyboard zoning.
4. Simple rhythmic seqs for illustrating external MIDI control options. With EXT TRIG on, can show aspects such as:
  - triggering seq from single notes & chords
  - building up multiple notes with BUILD on then collapsing back to 1

- control of seq velocity from keyboard
- behaviour with MOMentary on
- different seqs triggered from different pitches
- second level of transposition with TRANSPOSE on
- triggering several seqs simultaneously (set to flashing state then play a note on keyboard)
- CYCLE behaviour (set several to flashing state then play several notes)
- control from different zones with GROUP on
- control & transposition of drum parts (Control Keys 10 - 12)

5. Effects (Trigger all from single notes on keyboard):

- chords on Control Keys 1 & 2
- arpeggio effects on 3 - 6
- hold-at-end effects on 7 - 10

6. Effects (Trigger from notes & chords on keyboard. Try different tempos.)

- rhythms on 1 - 3. Try combining the rhythms.
- 'strumming' effect on 5 with Thru (for comparison) on 4
- 'chiff' effect on 7 with Thru (for comparison) on 6

7. Blank

8. Blank

9. Blank

10. Drum patterns. Some left blank for experimentation.

11. Config to illustrate use of GROUP, CYCLE & single-shot sequences.

12. Big orchestral chord. Play from single notes on keyboard.

13. General-purpose.

14. General-purpose.

15. 'Fanfare'. Useful for showing how sequences can be triggered against themselves at different times.

16. 'Spanish'. Useful for showing STEP option.

17 - 24. Unused.



**Doc: 'HAE'**

KAL: USING HOLD-AT-END SEQS IN DEMO CART 28/2/88

-----

There are 6 HAE examples in Config 20. Use these 1 at a time, playing from single notes on the keyboard - the octave below middle C is best. Show effect of keyboard velocity & of changing tempo.

For comparison you have Thru to the same sounds on Control Keys 7 - 12. This lets you show what the MPS is contributing to the sound.

Note: All these sounds are on the MT32. It sometimes gets confused with its 'intelligent' voice-assignment - just play the HAE effect a couple of times and it sorts itself out.

#### THE 'QUANTISATION BUG'

-----

I think this happens when you record in real time with RESTART off - it's very easy to get out of sync in this case. Normally RESTART goes on automatically when you go into real-time record, but it doesn't if you have a Trigger Profile defined in the current Config.

I'm altering this in software to make it more user-friendly. In the meantime, always check that RESTART is on and RQ + ALIGN are off when recording in real time.